Varnika Bajpai

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Education

KJ Somaiya School of Engineering, BTech, Information Technology with Honours in Artificial Intelligence 2022 – 2026

- CGPA: 9.66. Silver Medallist, Information Technology.
- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Machine Learning, Deep Learning, NLP

Skills

- **Languages & Frameworks** – Python, C++, Java, JavaScript (ES6+), React (React.js), Node.js, Django, Flask, FastAPI.
- **Data, ML & Analytics** – SQL, NoSQL, Pandas, Scikit-learn, NumPy, XGBoost, LLMs (RAG), Grafana, Tableau.
- **Cloud & DevOps** – Google Cloud Platform (Certified), AWS, Docker, Git, GitLab CI/CD, RESTful APIs.

Experience

Software Engineer, Barclays – Pune, MH July 2026 – Present

- Build engineering tooling and automation in Python, with Machine Learning (ML) development and MLOps.

Digital Technology Intern, SAP Testing, LyondellBasell – Mumbai, MH January 2026 – June 2026

- Built a backend integration pipeline for testing analytics and visualisation with Python and REST APIs.
- Traversed deep parent-child hierarchies in the test management system with tree traversals.
- Scaled it across enterprise projects on an AWS server with near real-time scheduled runs, feeding self-refreshing dashboards with zero manual steps.

Technology Summer Intern, Barclays – Pune, MH May 2025 – July 2025

- Delivered internal automation and developer tooling, cutting manual triage time by about 40% in pilot.

Summer Intern, G-Square Solutions (AI/ML and BI Solutions Provider) – Mumbai, MH June 2024 – July 2024

- Developed backend ML models using Django and Flask frameworks trained on 1,500+ samples and shipped endpoints used across product modules.
- Created SQL views and integrated dashboards, reducing refresh time from 2–3 hours to minutes.

Projects

OmniCompiler *React, FastAPI, Docker, Python, Gemini API, Scikit-learn (Random Forest)*

- Built a language-agnostic platform to run, debug, translate and analyse code in Python, JavaScript, Java, C++ and Go, with sandboxed Docker runtimes and control-flow graphs.
- Developed the breakpoint recommendation end to end: one Random Forest model per language, trained on features of each line of code, that suggests where to pause.

FloatChat *React, Node.js, Express, MongoDB, Python, PostgreSQL, Chroma, GPT-5 API*

- Designed a system enabling natural language queries over ocean datasets using a custom MCP(Model Context Protocol) and Retrieval-Augmented Generation (RAG).
- Developed an assistant to query, analyze, and visualize Argo oceanographic data through conversational prompts, returning insights grounded in real NetCDF files.

Symbiote *React, Node.js, MongoDB, Python (ML), Socket.io*

- Built an AI-powered platform to match students into balanced hackathon teams using skills, GitHub, and EQ traits.
- Enabled real-time team formation, chat, and role setup via intelligent matching and Socket-based collaboration tools.

MalShield *Python, LightGBM, YARA*

- Created an ML-powered malware detection tool using static and dynamic analysis and YARA rules (EXE, DOC, BAT).
- Incorporated multi-format parsing and embedded ML pipelines to detect, classify, and log malicious behavior efficiently.

Achievements and Volunteer Work

- **Idea Hackathon - UBI Bharosa**
  - Engineered ML models (ArcFace, XGBoost, Random Forest) with up to 97% facial recognition accuracy and 90%+ precision in customer prioritization for UBI Bharosa, an AI-powered scheduler and smart ticketing system.
  - Secured 3rd place out of 500+ teams in Union Bank of India's Idea Hackathon (PSB Series).
- **Computer Society of India** – Organized national hackathons (Smart India Hackathon, Devopia) and gave AI/ML talks.
- **Somaiya Debating Society** – Engaged students and presided over Big Somaiya Debate, ensuring fair competition.

Certificates

React - The Complete Guide (Udemy), Generative AI Basics & NLP (Salesforce Trailhead), Google Analytics for Beginners